(f) $\underbrace{\sum_{n\geq 1} \sin\left(\pi\sqrt{n^2+1}\right)}_{=(-1)^n x_n \to 0}$. 0.5p

$$\sin(x - n\pi) \begin{cases} \sin(x - 2k\pi) = \sin(x) \\ \sin(x-(2k+1)\pi) = \sin(x-\pi) = -\sin x. \end{cases}$$

$$\sin(x - n\pi) = (-1)^n \sin(x).$$

$$\sin\left(\pi\sqrt{n^2+1} - n\pi\right) = \sin\left(\pi\sqrt{n^2+1} - \pi\sqrt{n^2}\right) = \sin\left(\pi \cdot \frac{1}{\sqrt{n^2+1}+\sqrt{n^2}}\right).$$

$$\sin\left(\pi\sqrt{n^2+1}\right) = (-1)^n \sin\left(\pi\sqrt{n^2+1} - n\pi\right) = (-1)^n \cdot \underbrace{\sin\left(\frac{\pi}{\sqrt{n^2+1}+n}\right)}_{=: x_n}.$$

$$x_n \searrow 0 \overset{\text{Leibniz}}{\Longrightarrow} \sum (-1)^n x_n \text{ convergent.}$$

- **Theorem:** If $\sum_{n=1}^{\infty} x_n$ converges, then $\lim_{n\to\infty} x_n = 0$.

3. Find the MacLaurin series and its radius of convergence for the following functions:

(c) $\sin^2(x)$.

(d) $\arctan x$.

MacLaurin Series: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$.

c) $f(x) = \sin^2(x)$.

Method 1: Compute $f^{(n)}(0)$, then write $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

Method 2: Write $f(x)$ in a different way, and use the Taylor series for other functions.

Method 1:

$f'(x) = 2\sin(x)\cos(x) = \sin(2x)$, $f'(0) = 0$

$f''(x) = 2\cos(2x)$, $f''(0) = 2$.

$f^{(3)}(x) = -4\sin(2x)$, $f^{(3)}(0) = 0$

$f^{(4)}(x) = -8\cos(2x)$, $f^{(4)}(0) = -8$.

$f^{(2n+1)}(0) = 0$, $f^{(2n)}(0) = (-1)^{n+1} \cdot 2^{2n-1}$, $n \geq 1$.

$$f(x) = \underbrace{f(0)}_{=0} + \underbrace{f'(0)}_{=0} \cdot x + \sum_{n=1}^{\infty} \underbrace{\boxed{\frac{(-1)^{n+1} 2^{2n-1}}{(2n)!}}}_{a_{2n}} x^{2n}$$

$R = \frac{1}{L}$, $L = \lim \frac{a_{2n+2}}{a_{2n}}$
$=\infty$ $\qquad = 0$

Method 2: $\sin^2 x = \frac{1-\cos 2x}{2}$.

$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$

We can find the MacLaurin series for $\cos 2x$.

Use $\cos t = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} + \dots$ and take $t = 2x \dots$

↳ Taylor Series for $\cos t$, valid for $\forall t \in \mathbb{R}$.

$$\cos 2x = \sum_{n=0}^{\infty} (-1)^n \frac{(2x)^{2n}}{(2n)!}, \qquad \frac{1-\cos 2x}{2} = \frac{1}{2} - \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 2^{2n-1} x^{2n}}{(2n)!}$$

$$= \frac{1}{2} + \sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^{2n-1} x^{2n}}{(2n)!}$$

$$= \frac{1}{2} - \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2^{2n-1} x^{2n}}{(2n)!}$$

$$= \frac{1}{2} - \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2^{2n-1} x^{2n}}{(2n)!}$$

- $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$

  $\sin^2 x = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right)^2 = ???$

  $(a_1 + a_2 + \ldots a_n + \ldots)^2 = a_1^2 + a_2^2 + \ldots + a_n^2 + 2\sum a_i a_j.$

- $\sum_{n=0}^{\infty} a_n x^n$ Radius of convergence $R = \frac{1}{L}$, $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$.

  Abs Conv. $\forall x \in (-R, R)$. $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$.

- $\sum_{n=0}^{\infty} a_n (x-c)^n$, Abs conv. $\forall x \in (c-R, c+R)$

  $c-R \quad c \quad c+R$

4. Find the sum and the convergence set for the power series $\sum_{n\geq 1} nx^n$. 1p

$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots = \frac{1}{1-x}$, $L = 1 \Rightarrow \boxed{R = 1}$, Abs Conv. $\forall x \in (-1, 1)$.

$(x^n)' = n x^{n-1}$

$$\left(\sum_{n=0}^{\infty} x^n\right)' = \sum_{n=0}^{\infty} (x^n)' = \sum_{n=1}^{\infty} n x^{n-1} = \left(\frac{1}{1-x}\right)' = \frac{1}{(1-x)^2}.$$

$$\Rightarrow \sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2} \;\Big|\cdot x \Rightarrow \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}.$$

Convergence Set: $(-1, 1)$ and we need to check $x = -1$, $x = 1$.

- $x = 1$: $\sum_{n=1}^{\infty} n = \infty$, divergent, • $x = -1$: $\sum_{n=1}^{\infty} n(-1)^n$ divergent.

Convergence Set $= (-1, 1)$.

$a_n = \frac{1}{n(n-1)}$, $L = 1$, $R = 1$.

2. Study the convergence and compute the sum for the series $\sum_{n\geq 2} \frac{x^n}{n(n-1)}$. $= A + B = x\ln(1-x) - \ln(1-x) - x$.

$$\frac{x^n}{n(n-1)} = x^n \cdot \frac{1}{n(n-1)} = x^n \left(\frac{1}{n-1} - \frac{1}{n}\right) = \frac{x^n}{n-1} - \frac{x^n}{n} = x \cdot \frac{x^{n-1}}{n-1} - \frac{x^n}{n}.$$

? $\sum_{n=2}^{\infty} \frac{x^n}{n} = A$ and $\sum_{n=2}^{\infty} x \frac{x^{n-1}}{n-1} = x \sum_{n=1}^{\infty} \frac{x^n}{n} = B$

$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots = \frac{1}{1-x}$, $\forall x \in (-1, 1)$. | Integrate

$\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = -\ln(1-x)$, $\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x) \Rightarrow B = -x\ln(1-x)$

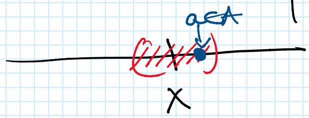
$A = -\ln(1-x) - x$.

$[\ln(1-x)]' = \frac{(1-x)'}{1-x} = \frac{-1}{1-x}.$

- Radius of convergence $R = \frac{1}{L} = 1$, Abs Conv. $\forall x \in (-1,1)$.
- $x=1$: $\sum_{n=2}^{\infty} \frac{1}{n(n-1)}$ "like" $\sum_{n=2}^{\infty} \frac{1}{n^2}$ convergent.
- $x=-1$: $\sum_{n=2}^{\infty} \frac{(-1)^n}{n(n-1)}$ converges by the Leibniz Test.

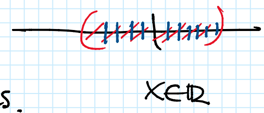
- Accumulation points: x is an acc point of A if:

$$\forall V \in \mathcal{V}(x),\ V \cap (A \setminus \{x\}) \neq \emptyset$$



Ex: Accumulation points of $\mathbb{Q}$.

Any $x \in \mathbb{R}$ is an accumulation point for $\mathbb{Q}$.
because any $V \in \mathcal{V}(x)$ contains rational numbers.



Can the averages converge, but the sequence not?

(b) Compare $1 + \frac{1}{2} + \ldots + \frac{1}{n}$ with n and $\ln n$ by taking the ratio, respectively.

$$\lim_{n\to\infty} \frac{1+\frac{1}{2}+\ldots+\frac{1}{n}}{n} \overset{S-C}{=} \lim_{n\to\infty} \frac{\frac{1}{n+1}}{1} = 0 \Rightarrow 1+\frac{1}{2}+\ldots+\frac{1}{n} \text{ increases slower that } n.$$

$$\lim_{n\to\infty} \frac{1+\frac{1}{2}+\ldots+\frac{1}{n}}{\ln n} \overset{S-C}{=} \lim_{n\to\infty} \frac{\frac{1}{n+1}}{\ln \frac{n+1}{n}} = 1 \qquad n \ln \frac{n+1}{n} \to \ln e = 1$$

$\Rightarrow 1+\frac{1}{2}+\ldots+\frac{1}{n}$ has the same growth rate as $\ln n$.
$" = \mathcal{O}(\ln n)$

(c) $\sum_{n\geq 1} \frac{1}{n\sqrt[n]{n}}$. **0.5p**

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{conv. if } p>1 \\ \text{div if } p\leq 1 \end{cases}$$

$\boxed{\text{Don't}}$ take: $p = 1 + \frac{1}{n} > 1$.

Wrong Solution: $\sum \frac{1}{n\sqrt[n]{n}} = \sum \frac{1}{n \cdot n^{\frac{1}{n}}} = \sum \frac{1}{n^{1+\frac{1}{n}}}$, $p = 1+\frac{1}{n} > 1$.
$\Rightarrow$ convergence.

Correct: $\frac{n\sqrt[n]{n}}{n} = \sqrt[n]{n} \to 1$. $\qquad \sqrt[n]{n} = n^{\frac{1}{n}} = e^{\frac{1}{n}\ln n} \to e^0 = 1$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n\sqrt[n]{n}}$ "like" $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$, divergent.

- Find the sum

(f) $\sum_{n\geq 1} \frac{1}{\sqrt[3]{n}}$. $= \sum_{n=1}^{\infty} \frac{1}{n^{1/3}}$ divergent $\frac{1}{3} < 1$. $\sum \frac{1}{n^p} \begin{cases} \text{conv if } p>1 \\ \text{div if } p\leq 1 \end{cases}$
$\Downarrow$
sum $= \infty$.

$\sqrt[3]{n} < \ldots \quad \frac{1}{\ldots} > \frac{1}{\ldots}, \quad \sum^{\infty} \frac{1}{\ldots} \quad \sum^{\infty} \frac{1}{\ldots}$

sum $= \infty$.

$$\sqrt[3]{n} < n, \quad \frac{1}{\sqrt[3]{n}} > \frac{1}{n} \Longrightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}} > \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

(d) $\sum_{n\geq 1} n! \sin x \sin \frac{x}{2} \dots \sin \frac{x}{n}$, $x \in (0, \pi)$.

Ratio Test
$$\frac{x_{n+1}}{x_n} = \frac{\cancel{(n+1)!}\, \cancel{\sin x} \cdot \cancel{\sin \frac{x}{2}} \cdots \cancel{\sin \frac{x}{n}} \cdot \sin \frac{x}{n+1}}{\cancel{n!}\, \cancel{\sin x} \cdot \cancel{\sin \frac{x}{2}} \cdots \cancel{\sin \frac{x}{n}}} = (n+1) \sin \frac{x}{n+1}$$

$$= \frac{\sin \frac{x}{n+1}}{\frac{x}{n+1}} \cdot x \longrightarrow x \begin{cases} \text{conv. if } x < 1 \\ \text{div. if } x > 1. \end{cases}$$

Study the case $x=1$: use the Raabe-Duhamel test: $\lim\limits_{n\to\infty} n\left(\frac{x_n}{x_{n+1}} - 1\right)$.

$$\lim_{n\to\infty} n \left(\frac{1}{(n+1)\sin\frac{1}{n+1}} - 1 \right) \underset{t = \frac{1}{n+1}}{=\!=} \lim_{t\to 0} \left(\frac{1}{t} - 1\right) \cdot \left(\frac{t}{\sin t} - 1\right)$$

$$= \lim_{t\to 0} \left(\frac{1}{t} - 1\right) \cdot \frac{t - \sin t}{t} = \lim_{t\to 0} \frac{t - \sin t}{t^2} - \lim_{t\to 0} \frac{t - \sin t}{t} \overset{l'H}{=} \lim_{t\to 0} \frac{1 - \cos t}{2t} - \underbrace{\lim_{t\to 0} (1 - \cos t)}_{=0}$$

$$\overset{l'H}{=} \lim_{t\to 0} \frac{\sin t}{2} = 0 \Longrightarrow \lim_{n\to\infty} n\left(\frac{x_n}{x_{n+1}} - 1\right) = 0 < 1 \Longrightarrow \text{divergent series for } x = 1.$$

3. Find the sum of the following series:

(a) $\sum_{n\geq 1} \frac{1}{(n+\alpha)(n+\alpha+1)}$, $\alpha \in (0,1)$. **1p**

"like" $\sum_{n=1}^{\infty} \frac{1}{n^2}$ convergent.

$$\frac{1}{(n+\alpha)(n+\alpha+1)} = \frac{1}{n+\alpha} - \frac{1}{n+\alpha+1}. \qquad \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

• Telescoping series $\sum_{n=1}^{\infty} \frac{1}{(n+\alpha)(n+\alpha+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n+\alpha} - \frac{1}{n+\alpha+1}\right) = \frac{1}{1+\alpha} - \cancel{\frac{1}{2+\alpha}} + \cancel{\frac{1}{2+\alpha}} - \cancel{\frac{1}{3+\alpha}} + \dots + \cancel{\frac{1}{n+\alpha}} - \frac{1}{n+\alpha+1} + \dots$

$$\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n+\alpha} - \frac{1}{n+\alpha+1}\right) = \lim_{N\to\infty} \left(\frac{1}{1+\alpha} - \frac{1}{N+\alpha+1}\right) = \lim_{n\to\infty} \left(\frac{1}{1+\alpha} - \underbrace{\frac{1}{n+\alpha+1}}_{\to 0}\right) = \frac{1}{1+\alpha}.$$

$$= \frac{1}{1+\alpha}.$$

(b) $\sum \frac{n(n+1)}{2^n}$. **1p** Find the sum

∞

(b) $\sum_{n\geq 2} \frac{n(n+1)}{2^n}$. **1p** $= f(\frac{1}{2})$

Find the sum

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

Find $\sum_{n=2}^{\infty} n(n+1)\, x^n$ and then take $x = \frac{1}{2}$.

$$\sum_{n=2}^{\infty} n(n+1)x^n = x \sum_{n=2}^{\infty} \underbrace{n(n+1)\, x^{n-1}} = x \cdot \sum_{n=2}^{\infty} (x^{n+1})''$$

$$(x^{n+1})' = (n+1)\, x^n$$

$$(x^{n+1})'' = n(n+1)\, x^{n-1}.$$

$$= x \cdot \left(\sum_{n=2}^{\infty} x^{n+1}\right)'' = x \cdot \left(\frac{1}{1-x} - 1 - x\right)'' = x \cdot \left(\frac{1}{1-x}\right)'' = f(x).$$